

Appendix G. Exploratory analysis: GEDI as the temporal LiDAR dimension

(Evidence base for the General Discussion perspectives. Exploratory status: single summer of ALS truth, microclimate tests at Blois only. All scripts and tables in NC_Full/chapter4_GEDI/ and the simulation repository; ~30 min of compute end to end.)

G.1 Data and processing

We used the GEDI L2A/L2B footprints already extracted over the three sites (9190 footprints, 2019–2023 acquisitions restricted here to 2021–2022: Aigoual 516, Blois 2448, Mormal 6226), each carrying relative-height metrics, total plant area index (PAI), the vertical PAI profile in 5 m bins (Dubayah et al., 2020), geolocation-corrected coordinates, and ALS LAI/LAD extracted at the corrected footprint position ($k = 0.5$, rescaled by 0.769 to the study convention $k = 0.65$). Quality filtering follows the literature and our own sweep: full-power beams only, sensitivity ≥ 0.90 , PAI > 0 , ALS LAI $\geq 2 \text{ m}^2 \text{ m}^{-2}$. The beam filter is the single most effective lever: at Blois the footprint-level correlation with ALS LAI rises from $r = 0.49$ (all beams) to $r = 0.67$ (power beams, $n = 635$). Because successive GEDI dates sample different footprints, all temporal composites are adjusted to a constant structure by regressing PAI on acquisition date and ALS LAI, and reported at the common mean ALS LAI; a common-support control gives the same curves. Phenological metrics use the 50% seasonal-amplitude threshold (TRS50) on weekly-interpolated, Whittaker-smoothed series (Atzberger & Eilers, 2011), the metric of Cotrina-Sanchez et al. (2026); the Elmore double-logistic misbehaves on one pooled season of sparse points and is kept only as a control.

Microclimate tests reuse the Chapter 3 machinery unchanged: MuSICA (Ogée et al., 2003) at the 53 HOBO plots of Blois, v3.2.3-iter binary, per-plot wind correction, total (two-sided) leaf area, scenarios validated against observed ΔT_{max} and the hourly micro-macro slope.

G.2 A waveform seasonal cycle that ranks canopy density

Monthly composites at Mormal (10 months sampled) trace a full deciduous cycle: winter floor $0.5\text{--}0.8 \text{ m}^2 \text{ m}^{-2}$ (December–March, wood and branch area), leaf-out by May, summer plateau $3.4\text{--}3.6$, decline through October (Fig. G1). The seasonal amplitude, a leaf-only proxy since the wood floor subtracts out, increases monotonically across ALS-LAI strata ($2.3 / 3.1 / 3.3 \text{ m}^2 \text{ m}^{-2}$ for strata 2–3 / 3–4 / >4), and the summer plateau ranks density at both sites (Fig. G2). This is the temporal counterpart of the Chapter 2 result: where the optical signal saturates, the waveform amplitude still discriminates.

G.3 Timing: leaf-out confirmed, senescence arbitrated

At Blois, per-acquisition composites adjusted to constant structure (nine acquisitions forming eight composite points, 2021 and 2022; Fig. G3) yield three timing results. (i)

A waveform-LiDAR seasonal cycle: GEDI monthly composites

Purple: GEDI power-beam median (IQR ribbon; dashed = structure-adjusted). Orange (Blois): S2 ATBD annual series. Winter floor = wood/branch area

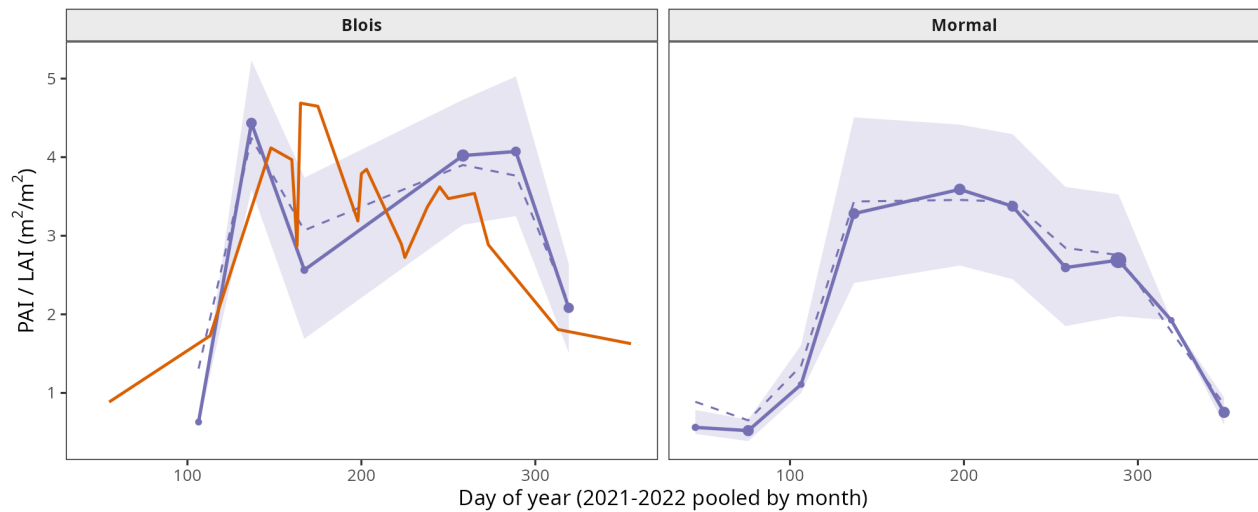


Figure 1: Figure G1. GEDI monthly composites (power beams, structure-adjusted; ribbon = IQR; dashed = adjusted). Orange: the Sentinel-2 ATBD annual series at Blois. Winter floor = wood/branch area.

Does the GEDI seasonal amplitude rank canopy density?

Winter = Dec-Mar (Mormal) / leaf-off proxy; Summer = Jun-Sep; power beams

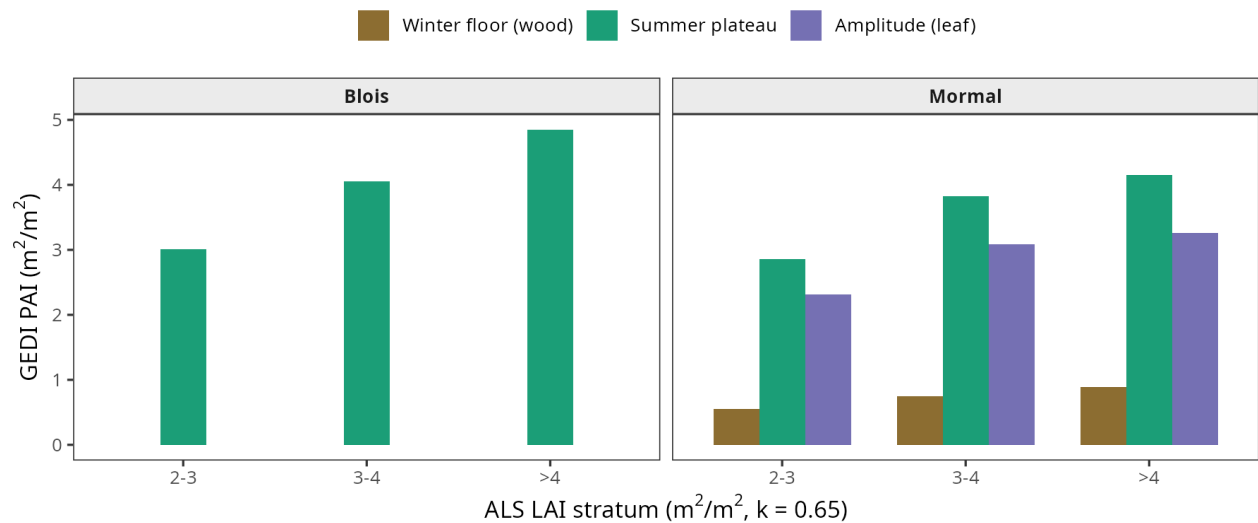


Figure 2: Figure G2. Winter floor, summer plateau, and seasonal amplitude by ALS-LAI stratum. The amplitude ranks canopy density where the summer optical signal is flat.

The 2021 summer plateau was not flat: June 6 sat $0.68 \pm 0.41 \text{ m}^2 \text{ m}^{-2}$ below September ($z = -3.2$), and GEDI lands on the Sentinel-2 June values, confirming that the S2 dip reflected the late, cold spring of 2021 rather than inversion noise. (ii) The 2022 drought summer left no canopy signature (September 2022 slightly above September 2021 at matched structure), while May 14, 2022 was already at seasonal maximum: GEDI resolves interannual phenology that a single-summer ALS snapshot cannot. (iii) TRS50 phenometrics (Table G1) put the GEDI start of season within three days of Sentinel-2 (122 vs 125), but the end of season about 25 days later (34 days with the unsmoothed spline metric of the lab notebook), replicating at site level the ~ 35 -day continental gap of Cotrina-Sanchez et al. (2026). Layer-resolved composites (Fig. G4), transposing the method of Oliveira & Zhang (2025), show the understory (0–10 m) greening up about seven days before the overstory (TRS50 116 vs 123), a measured vernal window.

Blois: per-acquisition GEDI composites, structure-adjusted

Points: GEDI power beams, adjusted to common ALS LAI (95% CI). Orange: S2 ATBD annual series (2021).

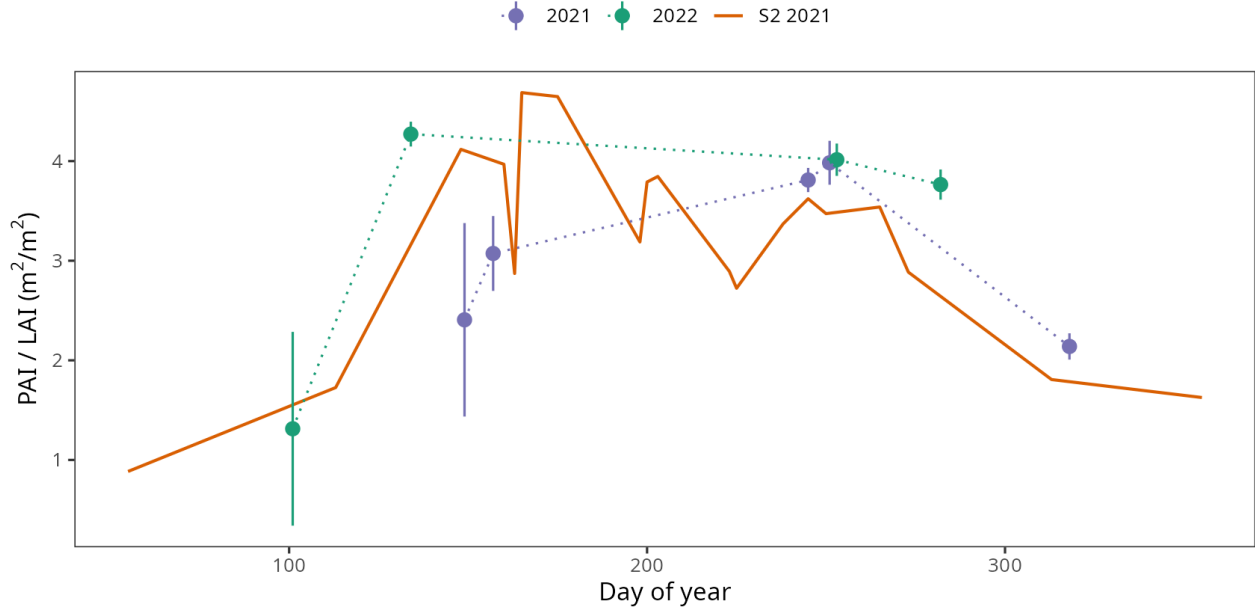


Figure 3: Figure G3. Blois per-acquisition GEDI composites, adjusted to a common ALS LAI (95% CI), against the Sentinel-2 2021 series.

Table G1. TRS50 phenometrics (day of year; weekly-interpolated, Whittaker-smoothed series).

Series	SOS	EOS
Blois, Sentinel-2 ATBD series (19 dates, 2021)	125	279
Blois, GEDI adjusted composites (eight composite points)	122	304
Mormal, GEDI total (10 months)	123	305
Mormal, GEDI understory (0–10 m)	116	298
Mormal, GEDI upper canopy (>10 m)	123	312

Two robustness observations complete the timing picture. The phenometrics transfer

Layer-resolved GEDI phenology (Oliveira & Zhang method, ALS-adjusted)

Power beams, adjusted to common ALS LAI. Does the understory lead in spring and lag in autumn?

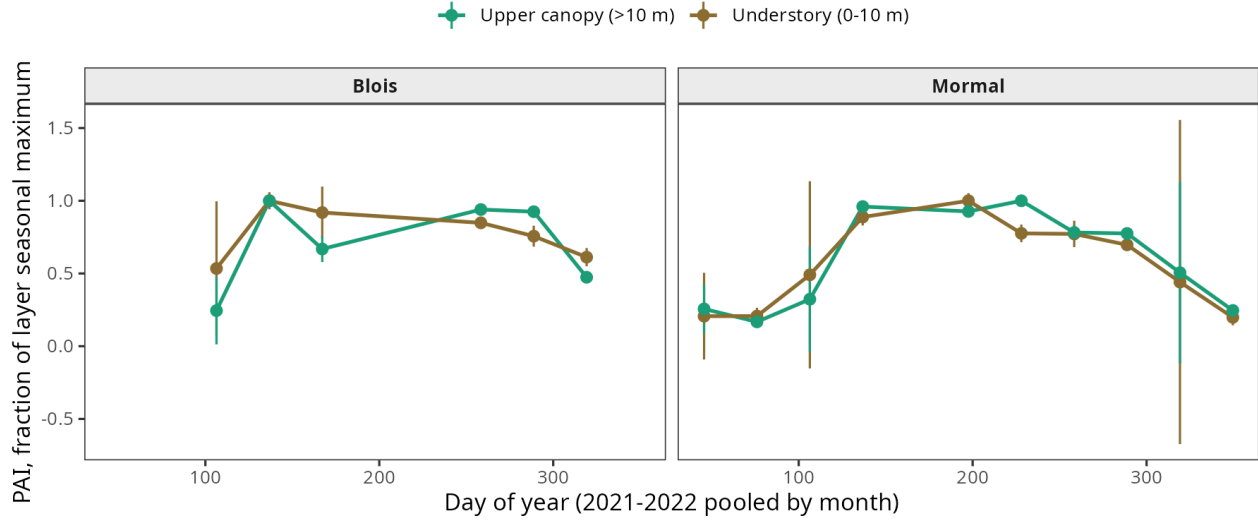


Figure 4: Figure G4. Layer-resolved GEDI phenology (understory 0–10 m vs upper canopy), structure-adjusted monthly composites normalized by each layer’s seasonal maximum.

across sites: SOS 122 vs 123 and EOS 304 vs 305 between Blois and Mormal (~250 km apart), although the full leaf-fraction curves overlap only moderately ($r = 0.74$, maximum difference 0.28), because the Blois curve carries the real 2021 spring anomaly that the Mormal months do not sample; the shoulders transfer, the mid-season carries local interannual signal. And at Mormal the TRS50 end of season is invariant across canopy-density strata (305 / 305 / 307 for ALS-LAI 2–3 / 3–4 / >4) while the start of season comes about 12 days earlier in the densest stratum (111 vs 123), an observation to read with species-composition caveats but directly relevant to the structure-phenology-microclimate coupling of Wu et al. (2024).

G.4 The arbitration matters for microclimate

We cloned the Chapter 3 seasonal scenario (LiDAR magnitude $\times$ Sentinel-2 timing) and replaced only the autumn limb (day of year ≥ 250) with the GEDI-measured leaf fraction. Against the 53 loggers, the GEDI-timed autumn significantly improved both metrics over the S2-timed series (paired bootstrap, 47 plots: $\Delta\text{RMSE} -0.083^\circ\text{C}$ [-0.105 , -0.060]; $\Delta R^2 +0.188$ [$+0.064$, $+0.404$]), turning the series’ worst window (autumn R^2 0.10, below a constant-LAI control) into its best RMSE. A second, pre-registered-in-spirit correction, filling the summer dips of the S2 series up to the plateau while keeping the peak, added a smaller but significant summer gain (JJAS $\Delta\text{RMSE} -0.066$ [-0.085 , -0.048] vs the raw series). Month-by-month decomposition attributes the summer effect to September and August, confirms June 2021 as real signal, and flags the mid-summer S2 wobble as noise, exactly as the GEDI plateau suggested (Table G2).

Table G2. Paired-bootstrap contrasts against the S2-timed seasonal series (47 plots with a full series, $B = 3000$; ΔRMSE in $^\circ\text{C}$ on ΔT_{max} ; brackets = 95% CI; source tables

15-18 of the working folder).

Contrast	Window	Δ RMSE	ΔR^2
GEDI autumn limb	Oct-Nov	-0.083 [-0.105, -0.060]	+0.188 [+0.064, +0.404]
GEDI autumn limb	JJAS (control)	-0.021 [-0.032, -0.012]	+0.022 [+0.006, +0.044]
+ summer dip-filling	JJAS	-0.066 [-0.085, -0.048]	+0.035 [+0.014, +0.060]
+ summer dip-filling	August alone	-0.158 [-0.211, -0.109]	+0.273 [+0.135, +0.488]

G.5 One irreplaceable role per sensor

A 15-scenario matrix ($\{\text{ALS, S2, GEDI}\} \times \{\text{static, temporal}\}$, same machinery, Table G3) condenses the exploration.

Table G3. Sensor-combination matrix, summer (JJAS) and autumn, condensed to the informative rows (full 15-scenario table in the working folder, Table 20). Δ Tmax metrics vs 53 loggers; pooled summer R^2 inherits the open-canopy composition effect of Chapter 3 and is read jointly with RMSE and bias.

Combo (type)	Summer R^2 / RMSE / bias	Autumn R^2 / RMSE
S2 (static)	0.56 / 1.75 / 0.79	0.31 / 1.63
ALS (static)	0.24 / 1.76 / 0.73	0.35 / 1.63
ALS+S2, Ch3 layered fusion	0.50 / 1.66 / 0.79	0.34 / 1.63
ALS+S2 (temporal, S2-timed)	0.24 / 1.76 / 0.73	0.10 / 1.66
ALS+GED I (temporal, GEDI shape)	0.15 / 1.91 / 0.89	0.30 / 1.58
S2+GED I, no ALS (temporal)	0.18 / 1.84 / 0.84	0.33 / 1.63
GED I only (temporal)	0.01 / 2.13 / 1.16	0.29 / 1.64
ALS+S2+GED I (temporal, corrected)	0.27 / 1.72 / 0.68	0.30 / 1.60

All low-bias scenarios carry the ALS magnitude; every temporal variant where GEDI supplies the per-plot magnitude fails in summer ($R^2 \leq 0.18$), and the two static GEDI-magnitude scenarios score higher (0.32 direct at footprints, 0.49 ratio-scaled Sentinel-2) but carry the largest warm biases of the matrix (+1.08, +1.23 °C), because GEDI’s dense-canopy skill, strong at its footprints ($r = 0.67$; bottom-layer agreement $r = 0.84$ where Sentinel-2 is blind), does not regress beyond them through wall-to-wall features. A site-level GEDI shape cannot replace the plot-scale Sentinel-2 timing either (summer R^2 0.15). What GEDI adds, uniquely, is the seasonal arbitration: both GEDI-timed scenarios take the best autumn RMSE, and the triple combination (ALS magnitude +

S2 timing + GEDI correction) is the best balanced package (best summer bias 0.68 °C, second RMSE in both windows). For the 11-18 plots within 100-150 m of a power footprint, the mean PAI of the three nearest footprints, used directly, recovers 80-90% of the ALS microclimate correlation ($r = -0.70$ to -0.73 / -0.77 to -0.80 against -0.86 / -0.89 to -0.90 for plot ALS), decaying beyond 200 m at the pace of the forest's own spatial decorrelation.

The composition closes the exploration. Applying the GEDI temporal correction inside the Chapter 3 height-gated fusion (tall plots: ALS magnitude $\times$ GEDI-corrected series; short plots: unchanged) produced the best product tested in either window: summer RMSE 1.61 °C (previous best 1.66), autumn R^2 0.46 and RMSE 1.57, with significant paired gains over the Chapter 3 fusion in both windows (summer Δ RMSE -0.066 [-0.084 , -0.048], ΔR^2 $+0.034$ [$+0.013$, $+0.056$]; autumn Δ RMSE -0.082 [-0.104 , -0.058], ΔR^2 $+0.194$ [$+0.064$, $+0.406$]). The Chapter 3 spatial rule and the GEDI temporal arbitration are complementary and compose without interference.

G.6 The 2022 heat test: the forcing rule transfers to an extreme year

The HOBO network stayed active through the record June and July 2022 heatwaves, and the simulations of the scenario machinery run to late July 2022, so the whole validation transfers to an out-of-calibration extreme year at no cost (Table G4). Two results. First, the observed LAI-buffering coupling holds under extreme heat: $r(\text{slope, LAI}) = -0.88$ on the p90 hottest days of June-July 2022 (-0.88 on the p90 hottest days of 2021 as well; -0.92 over all of summer 2021), the mean micro-macro slope does not collapse (0.93 against 0.86 on the hottest 2021 days), and the mean offset on hot days is deeper (-1.13 °C on 2021 hot days, -1.22 °C on 2022 hot days, against -0.73 °C over all of summer 2021), as a sub-unity slope under hotter skies implies; the GEDI composites show the 2022 canopy itself intact (§G.3). The invariant coupling, not the deeper offset, is the evidence relevant to the microrefugia premise (De Frenne et al., 2019; Lenoir et al., 2017). Second, the scenario ranking transfers: the two fusion products dominate the 2022 window (ΔT_{max} R^2 0.40 and 0.39, RMSE 1.80-1.81 °C) while the static ALS scenario collapses (R^2 0.08). One honest nuance: the warm bias grows in the extreme year ($+1.16$ °C vs $+0.73$ °C in 2021), so the model under-buffers more when heat is most severe, a clear target for model improvement.

Table G4. Observed coupling and scenario validation in the 2022 heatwave window (June 1 - July 29, 2022; $n = 51$ plots; 2021 rows $n = 53$).

Quantity	2021 JJAS, all days	2021 JJAS, hot p90	2022 Jun-Jul, all days	2022 Jun-Jul, hot p90
Mean observed ΔT_{max} (°C)	-0.73	-1.13	-1.05	-1.22
$r(\text{slope, ALS LAI})$	-0.92	-0.88	-0.89	-0.88
Mean micro-macro slope	0.90	0.86	0.90	0.93

Scenario (2022 window)	$\Delta T_{\max} R^2$	RMSE (°C)	Bias (°C)
Ch3 fusion (FUSION_H)	0.40	1.81	+1.17
+ GEDI correction (FUSION_GEDIMAX)	0.39	1.80	+1.16
Sentinel-2 static	0.27	2.12	+1.41
ALS static	0.08	2.11	+1.32

G.7 The vernal window in degrees: which way the association points

Per-plot green-up dates were taken from the full-year Sentinel-2 series of each plot (the annual ATBD series of Chapter 3, 47 plots) with the smoothed half-amplitude threshold of Table G1 (weekly interpolation, Whittaker smoothing, last upward crossing before the seasonal peak); they fall between day of year 106 and 134. An earlier version of this test used the summer-only series and a raw first crossing, which dated half the plots in January; those numbers are withdrawn. Crossing the corrected dates with the plot's own observed spring temperature (mean daily maximum, March 15 - April 30, 2021; $n = 33$ plots with both) gives $r = +0.55$, and $+0.40$ after partialling out ALS LAI ($+0.33$ after LAI and maximum height). Plots that leaf out later stay open longer and accumulate a warmer understory spring. The reciprocal test (pre-season winter temperature, January 1 - March 14, against green-up, in the spirit of Wu et al. (2024); $n = 35$) gives $r = +0.40$ but $+0.19$ once ALS LAI is partialled out, so most of the winter association runs through leaf area. Green-up date is itself correlated with ALS LAI ($r = -0.67$, denser plots dated earlier), which is why the partial coefficients are the ones to read. Both tests are cross-sectional and share their confounds (April canopy openness, composition, topography); together they say that the spring understory temperature follows phenology more closely than winter temperature anticipates it, and they connect the GEDI understory lead of §G.3 to the energy balance it implies.

G.8 Ancillary logger analyses (night, and how many loggers)

Two observation-only analyses of the 2021 logger archive complete the evidence base of the General Discussion. First, the night. Computing $\Delta T_{\min}$ (daily-minimum micro minus macro) alongside $\Delta T_{\max}$ over summer 2021 gives a near-mirror pair of couplings to LiDAR leaf area: $r = -0.89$ for $\Delta T_{\max}$ and $+0.83$ for $\Delta T_{\min}$ ($n = 53$). Closed canopy ($\text{LAI} > 4$) holds the night at the reference temperature (mean $\Delta T_{\min} +0.09$ °C) while open stands ($\text{LAI} < 1.5$) fall 2-4 °C below it (-3.15 °C), the mirror of the daytime pattern and a bilateral amplitude compression under a single structural control; the mean $\Delta T_{\min}$ is -0.66 °C with a between-plot SD of 1.24 °C, so the night field carries real structure. The macro minimum is taken from the MuSICA forcing file, so cold-air pooling at the reference is not excluded. This is a consistency check against the Chapter 1 attribution, not an attribution itself.

Second, network size. Bootstrap subsampling of the 53 loggers shows the observed slope-LAI coupling is establishable from small networks: at $n = 10$ the median r is -0.94 (95% CI -0.98 to -0.79), at $n = 20$ the CI tightens to $-0.96/-0.86$ and the regression coefficient is constrained to about ± 25 -30%, against $\pm 11\%$ at $n = 40$. Within this network, fifteen to twenty well-spread loggers reproduce the coupling; whether

that transfers to another site is untested, since the subsample inherits the site's own gradient.

G.9 Limits

Seven caveats bound these perspectives. The microclimate tests are single-site (Blois) and the composites pool 2021–2022, so interannual phenology is both signal and confound. Aigoual is a stated failure case for the footprint route: 516 footprints (203 matched), September only, relief and conifer admixture; footprint-level $r(\text{PAI}, \text{ALS LAI}) = 0.13$ (all beams) and 0.22 (power beams), and a GEDI-anchored magnitude correction over-corrects there (bias $-0.90 \text{ m}^2 \text{ m}^{-2}$). All microclimate results are Blois-only by construction. The late GEDI end of season mixes true leaves, marcescent leaves (typical of *Quercus* and *Fagus*), and exposed wood (Cotrina-Sanchez et al., 2026); the logger test suggests this structural signal is nevertheless the functionally relevant one for radiation and microclimate. Successive acquisitions sample different footprints, handled by structure adjustment, not by repeated measures. Several scenario variants were evaluated on the same logger network (47 series plots in the paired contrasts); the large effects (August, autumn) were pre-specified by the GEDI diagnosis, the small ones (July, September) were not and should be treated as suggestive. The 2022 test recycles the 2021 seasonal curves (the summer plateau makes this benign, and the GEDI composites support an unchanged 2022 canopy, but the leaf-out of 2022 was earlier and is not represented). And the vernal-window correlations (§G.7, $n = 33$ and 35) are cross-sectional: they order two associations, spring temperature after green-up and winter temperature before it, and do not identify a feedback.

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